

TRIX: Training-Free Reranking for Instruction-Following Retrieval with Exclusions

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Abstract

Retrieval requests often combine a target topic with positive criteria, background context, and explicit exclusions. Frozen retrievers rely heavily on lexical and semantic entity matching, overlooking how these instructions define relevance. This failure is acute for exclusions, where an excluded entity can become a strong matching signal. Existing remedies often require instruction training or representation adaptation. We introduce TRIX, a training-free reranking framework for frozen retrievers. TRIX preserves the complete request for retrieval and decomposes it into affirmative and exclusion views, focusing positive evidence and removing irrelevant background context. The two views can still share topic information, causing raw exclusion scores to penalize relevant content. TRIX removes the exclusion similarity predicted by affirmative relevance, then uses the residual to reduce positive scores and suppress exclusion matches. It requires no additional training data, model updates, or document re-encoding. Across three benchmarks and four frozen retrievers, TRIX improves retrieval effectiveness and instruction sensitivity, increasing the latter by up to 15.6 points and outperforming the strongest reported training-free baseline on retrieval with exclusion constraints by up to 3.3 points.

1 Introduction

Retrieval requests often combine a target topic with positive relevance criteria, background context, and explicit exclusions. Frozen retrievers rely mainly on lexical and semantic matching, but these signals do not reliably capture how each part of an instruction affects relevance. The problem is especially clear for exclusions: an excluded entity still appears in the request and can become a strong matching signal. Consider a request for evidence that a US college education improves employment or income while excluding documents about Manhattan colleges such as Columbia University. Documents about employment outcomes for Columbia graduates match the topic and entities in the request, yet they violate the exclusion. This behavior agrees with evidence that neural rankers are often insensitive to meaning changes introduced by negation (Weller, Lawrie, and Durme 2024; Weller et al. 2025b).

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Separating the instruction into positive and exclusion views is a natural way to make these roles explicit. This decomposition also exposes a second problem: the two views can still share topic information. A relevant document about a college outside Manhattan, for example, may score highly against the exclusion “Manhattan colleges such as Columbia University” because both concern US colleges and graduate outcomes. The exclusion score then reflects both shared topic information and evidence specific to the excluded content. Penalizing the complete score can therefore suppress documents that satisfy the request.

Existing work introduces instruction control through learned representations, inference-time query adaptation, or explicit constraint reasoning (Weller et al. 2025d; Zeng et al. 2026; Gao et al. 2026; Lee et al. 2026a; Xu et al. 2025; Hoveyda et al. 2026). Task-aware and instruction-tuned retrievers show that natural-language descriptions can condition retrieval across tasks and domains (Asai et al. 2023; Su et al. 2023; Cai et al. 2023). These approaches typically adapt the representation space or introduce an additional reasoning procedure. A complementary question is whether the scores of a frozen retriever can be corrected directly, without re-training the model or rebuilding the document index. Such a correction must retain affirmative relevance while distinguishing exclusion evidence from shared topical similarity.

We propose TRIX, a training-free reranking framework for this setting. TRIX retains the complete instruction for initial retrieval, derives an affirmative view that focuses the positive criteria and removes irrelevant background context, and represents excluded content in a separate view. For each request, it estimates the component of the exclusion score predicted by the affirmative score and uses the standardized residual as exclusion evidence. The final scoring rule preserves affirmative support when this evidence is weak, then reduces that support and applies an exclusion penalty as the residual increases. This design operates on the scores of a frozen retriever and requires no document re-encoding.

Across FollowIR, NegConstraint, and ExcluIR, TRIX improves instruction-aware ranking across sparse and dense frozen retrievers. It raises instruction sensitivity by up to 15.6 points and exceeds the strongest reported training-free reference on explicit-exclusion retrieval by up to 3.3 points. Our contributions are threefold: (i) we introduce a training-free IF-IR framework that uses full, affirmative, and exclu-

sion views to handle positive criteria, background context, and exclusions; (ii) we introduce residual correction to separate exclusion-specific evidence from shared topic information, together with a scoring rule that combines relevance promotion and exclusion suppression; and (iii) we provide controlled evidence across three benchmarks and multiple retriever families, separating the effects of evidence construction, residualization, and score composition.

2 Related Work

2.1 Learning Instruction-Following Retrievers

Instruction-following behavior can be learned through retriever adaptation. Early task-aware retrieval systems condition relevance on natural-language task descriptions: TART uses multitask instruction tuning, INSTRUCTOR learns a shared instruction-conditioned embedding space, and HypeR composes task-dependent prompts for cross-domain retrieval (Asai et al. 2023; Su et al. 2023; Cai et al. 2023). FollowIR-7B, Promptriever, and InF-IR train retrievers with natural-language relevance instructions, while Dual-View Training adds polarity-reversed supervision (Weller et al. 2025b,d; Zhuang et al. 2025; Zeng et al. 2026). InstEmb instead combines input-intrinsic and output-aware semantic signals when learning instruction-following embeddings (Gao et al. 2026). In each case, instruction sensitivity is encoded by the training data and objective. TRIX keeps the representation function fixed and introduces instruction control at the ranking stage.

2.2 Inference-Time Query Adaptation

Inference-time methods expose structure that a monolithic query may obscure. Query rewriting and expansion reformulate the request to improve relevance matching (Gao et al. 2023a; Wang, Yang, and Wei 2023a; Raudaschl 2024; Li et al. 2025). Stage-Aware Decomposition retains the complete query for initial retrieval and additively aggregates decomposed condition scores during reranking, treating each condition as positive matching evidence (Yin et al. 2026). DEO instead derives positive and negative subqueries and optimizes the query embedding toward the former and away from the latter (Lee et al. 2026a). TRIX also decomposes the request, but neither replaces nor optimizes its representation. It preserves the complete, affirmative, and exclusion queries as separate scoring views and resolves their evidence after the documents have been scored.

2.3 Constraint-Aware Ranking and Reasoning

Negation has long been a difficult case for learned representations. Targeted augmentation can improve negation sensitivity in pretrained language models (Singh, Kumar, and Sridhar 2023), while NevIR shows that many neural rankers struggle to distinguish otherwise similar documents whose meanings differ through negation (Weller, Lawrie, and Durme 2024). These studies motivate negation-aware modeling, but explicit exclusion retrieval additionally requires balancing a violation signal against the topical relevance shared by desired and excluded content.

Explicit constraints can be handled through score correction or structured reasoning. In vision-language retrieval,

SpaceVLM-DRC constructs multiple query directions and adds a repulsion penalty based on raw negated-concept similarity (Pokhrel and Sheshappanavar 2026). Directly penalizing raw exclusion similarity is unreliable in text retrieval when desired and excluded content share entities or topical context: a relevant document can score highly against both. TRIX instead leaves query and document representations unchanged, removes the component of the exclusion score predicted by affirmative relevance, and uses the remaining residual for reranking.

Structured approaches enforce constraints at a different inference level. NS-IR maps queries and documents into logical forms and enforces connective consistency for negative-constraint retrieval (Xu et al. 2025), whereas OrLog decomposes complex requests into predicates and applies probabilistic reasoning (Hoveyda et al. 2026). These methods target predicate-level constraint satisfaction, whereas TRIX provides score-level correction for frozen retrievers.

3 Methodology

TRIX has three steps. It retrieves candidates with the complete instruction, scores them under affirmative and exclusion views, and corrects their ranks using residual exclusion evidence. The following subsections define these evidence views, the residual exclusion score, and the asymmetric scoring rule.

3.1 Problem Formulation and Evidence Views

TRIX receives a natural-language request u containing the retrieval topic and its instructions. The frozen retriever uses the complete request as the full query $Q_{\text{full}} = u$ and returns the top- K candidate set $\mathcal{C}_u$. An LLM decomposes u once into two auxiliary queries: Q_{pos} states the content that supports relevance, and Q_{neg} states the content to exclude. If the request contains no exclusion, the decomposer returns $Q_{\text{neg}} = [\text{NONE}]$. The candidate set is always retrieved with Q_{full} ; the auxiliary views are used only for reranking.

The three views serve different roles. The full query retains the complete topic, context, and instruction during candidate generation. The affirmative view removes the exclusion clause and non-criterion context to provide a focused relevance signal. The exclusion view states the rejected content as a positive description, allowing the retriever to score whether a document matches that content. In addition to supporting relevance, the affirmative score provides a reference for the topical component shared with the exclusion score. All views use the same frozen scorer.

Let $s(Q, d)$ denote the score assigned to document d under query Q . For each $d \in \mathcal{C}_u$, the three channels are

$$\begin{aligned} S_{\text{full}}(d) &= s(Q_{\text{full}}, d), \\ S_{\text{pos}}(d) &= s(Q_{\text{pos}}, d), \\ S_{\text{neg}}(d) &= s(Q_{\text{neg}}, d). \end{aligned} \tag{1}$$

For a dense retriever, $s(Q_x, d) = \cos(f_q(Q_x), f_d(d))$ is the embedding similarity; for BM25, it is the lexical retrieval score. Dense document embeddings are computed once and reused across views, while BM25 uses one fixed index.

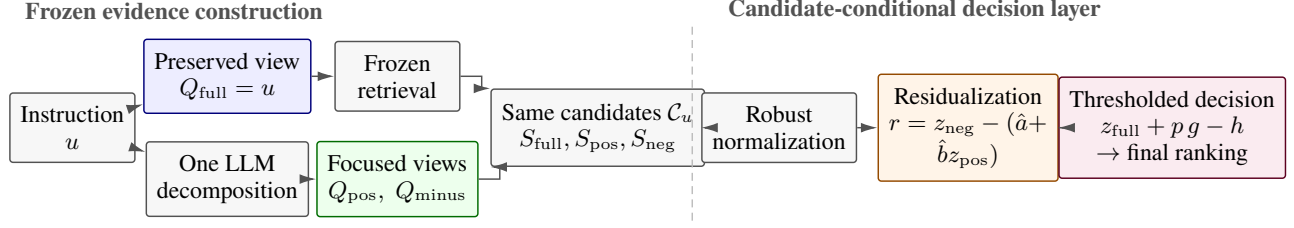


Figure 1: Overview of TRIX. The complete instruction is used for initial retrieval, while affirmative and exclusion queries score the retrieved documents with the same frozen retriever. TRIX obtains a residual exclusion score by robustly regressing exclusion scores on affirmative scores and uses the standardized residual in a thresholded reranking function.

Table 1 illustrates the decomposition. The affirmative view retains the desired content, and the exclusion view converts the rejected content into a directly scorable description.

View	Example text (Core17, qid 419)
Q_{full}	What new uses have been developed for old automobile tires as a means of tire recycling? A relevant document must show advantageous uses of recycled tires, such as destructive distillation of scrap rubber for valuable chemicals, reef building for fish habitats, filler or binder in asphalt roadway mixes, and burning in a controlled environment for heat generation. Exclude any references of rubber.
Q_{pos}	New uses for old automobile tires in recycling including destructive distillation, reef building, asphalt filler, and controlled burning for heat.
Q_{neg}	rubber

Table 1: Generated views for Core17 qid 419. Q_{full} is the complete request; Q_{pos} and Q_{neg} are produced by one LLM decomposition.

3.2 Residual Exclusion Score

The raw exclusion score mixes exclusion-specific evidence with topical similarity shared by the affirmative and exclusion views. For example, when the desired topic is genetically engineered food and the exclusion is the Flavr Savr tomato, a relevant document about genetically engineered soybeans can score highly under both views because both concern the broader topic. Directly subtracting the exclusion score penalizes this shared topical component together with evidence specific to the excluded content.

TRIX separates the two components by measuring whether a document’s exclusion score is unusually high given its affirmative score. For each request, it fits an affine relation between the normalized affirmative and exclusion scores over C_u . The standardized signed deviation from this relation defines the residual exclusion score. Fitting the relation per request adapts the reference to the score geometry induced by each instruction.

TRIX first robustly standardizes all three score channels

over the retrieved set C_u :

$$z_x(d) = \frac{S_x(d) - \text{median}_{d' \in C_u} S_x(d')}{\text{MAD}_{d' \in C_u}(S_x(d')) + \epsilon}, \quad (2)$$

$x \in \{\text{full}, \text{pos}, \text{neg}\},$

where $\text{MAD}(S) = \text{median}|S - \text{median}(S)|$ and $\epsilon > 0$ prevents division by zero. This normalization aligns query-specific offsets and scales while limiting the influence of extreme scores (Rousseeuw and Croux 1993). TRIX then fits the affine relation with Huber regression:

$$(\hat{a}_u, \hat{b}_u) = \arg \min_{a, b} \sum_{d \in C_u} \rho_\delta(z_{\text{neg}}(d) - a - b z_{\text{pos}}(d)), \quad (3)$$

where ρ_δ is the Huber loss with transition point δ . The fit captures the dominant score association, while the Huber loss limits the influence of large deviations (Huber 1964). It provides a query-specific score-space reference for exclusion similarity. For $e(d) = z_{\text{neg}}(d) - \hat{a}_u - \hat{b}_u z_{\text{pos}}(d)$, we robustly standardize the residual:

$$r(d) = \frac{e(d) - \text{median}_{d' \in C_u} e(d')}{\text{MAD}_{d' \in C_u}(e(d')) + \epsilon}. \quad (4)$$

The second standardization makes residual magnitudes comparable across requests: $r(d) = 0$ is the median fitted deviation, and one unit is measured relative to the within-request residual dispersion. This common scale allows the same decision parameters to be used across queries and retrievers. Positive $r(d)$ indicates more exclusion similarity than predicted by the document’s affirmative score; negative $r(d)$ indicates less. The residual is a relative score-space signal, not a calibrated probability of constraint violation. Numerical details are given in Appendix E.

3.3 Residual-Guided Asymmetric Scoring

TRIX uses the residual signal to regulate affirmative support and suppress strong exclusion matches. We first retain the positive part of the normalized affirmative score:

$$p(d) = [z_{\text{pos}}(d)]_+. \quad (5)$$

Because $z_{\text{pos}}(d)$ is median-centered, $p(d)$ rewards only documents whose affirmative score is above the candidate-set median. We activate the exclusion correction beyond a shared boundary $\lambda \geq 0$:

$$h(d) = [r(d) - \lambda]_+, \quad g(d) = \exp\left(-\frac{h(d)}{\tau}\right), \quad \tau > 0, \quad (6)$$

where $h(d)$ is the active exclusion evidence and $g(d) \in (0, 1]$ is its attenuation factor. Below the boundary, $h(d) = 0$ and $g(d) = 1$; above it, $h(d)$ grows linearly while $g(d)$ decreases smoothly. The final score is

$$S_{\text{final}}(d) = z_{\text{full}}(d) + p(d)g(d) - h(d). \quad (7)$$

The term $p(d)g(d)$ retains the affirmative reward under weak exclusion evidence and attenuates it once the boundary is crossed. The term $-h(d)$ directly suppresses strong exclusion matches, so the correction can lower a document below its full-query score. For fixed $z_{\text{full}}(d)$ and $p(d)$, Eq. 7 is therefore non-increasing in $r(d)$. When no exclusion view is present, TRIX sets $h(d) = 0$ and $g(d) = 1$, reducing the score to $z_{\text{full}}(d) + p(d)$.

4 Experiments

4.1 Experimental Setup

Benchmarks and Metrics. FollowIR (Weller et al. 2025a) is our primary benchmark for changed-instruction retrieval on Core17, Robust04, and News21. We report MAP on Core17 and Robust04, nDCG on News21, and p-MRR (Weller et al. 2025a) for sensitivity to instruction changes. We interpret relevance and p-MRR jointly because either can improve at the expense of the other. We also test transfer on NegConstraint (Xu et al. 2025), using MAP and nDCG@10, and ExcluIR (Zhang et al. 2025), using MRR and Right Rank (RR). RR is the percentage of desired–excluded document pairs ordered correctly. ExcluIR is used only for evaluation.

Models and Evaluation Protocol. The frozen Qwen3-4B model (Yang et al. 2025) produces Q_{pos} and Q_{neg} in one structured generation; Q_{full} remains the original request and retrieves the candidates. FollowIR experiments use BM25 (Robertson and Zaragoza 2009) together with the frozen E5-Mistral, BGE-large, and RepLLaMA bi-encoders (Wang et al. 2024; Xiao et al. 2024; Ma et al. 2024) and rerank each backbone’s top- K candidates, where $K = \text{TBD}$. NegConstraint and ExcluIR use BGE-small-en-v1.5, BGE-large-en-v1.5, and BGE-M3 (Xiao et al. 2024). Within every matched comparison, variants share candidates, generated views, retrieval templates, scorer configuration, and document representations. The same $\lambda = \text{TBD}$ and $\tau = \text{TBD}$ are fixed without test relevance labels and used across datasets and retrievers. Prompt, decoding, normalization, and numerical details are provided in Appendix E.

Baselines and Controlled Comparisons. On FollowIR, FollowIR-7B, Promptriever, and InstEmb (Weller et al. 2025a,c; Gao et al. 2026) provide instruction-trained system-level references. We also reproduce HyDE, Query2Doc, and ConvSearch-R1 with RepLLaMA (Gao et al. 2023b; Wang, Yang, and Wei 2023b; Jiang et al. 2025). Per-collection comparisons are reported in Appendix A. DEO (Lee et al. 2026b) and the strongest results reported with NegConstraint and ExcluIR provide system-level context. We report these separately from the matched Base/+*TRIX* comparisons because their training and retrieval pipelines differ. Paired uncertainty for the central component comparisons is reported in Appendix B.2.

4.2 Main Results on Instruction-Following Retrieval

Table 2 tests whether TRIX improves both relevance and instruction sensitivity across frozen retrievers. The lower block contains matched Base/+*TRIX* pairs spanning BM25 and three dense retriever families; the upper block provides instruction-trained system-level references.

The matched comparisons show that retrieval effectiveness alone does not imply instruction sensitivity. Before reranking, three dense backbones have negative average p-MRR despite competitive relevance scores. Adding TRIX makes average p-MRR positive for all four backbones and also raises their average relevance scores. The gains hold from BM25 to three dense retrievers: for RepLLaMA, TRIX raises the average Score from 23.8 to 28.4 and p-MRR from -3.3 to $+12.3$. Because the matched rows keep retrieval outputs and document representations fixed, these comparisons isolate reranking within each backbone’s candidate set.

Comparison with Query Rewriting and Expansion. The RepLLaMA comparison in Appendix A separates topical expansion from instruction-aware reranking. Query2Doc obtains the strongest average relevance score among the representative rewriting pipelines (26.2) but has negative p-MRR (-1.1); TRIX reaches 28.4 and $+12.3$, respectively. This pipeline-level comparison indicates that adding relevant query content is insufficient to ensure responsiveness to changed relevance criteria. The matched ablations below isolate the contribution of TRIX’s scoring components.

4.3 Transfer Across Exclusion Benchmarks

Table 3 tests whether the same reranking rule transfers from FollowIR to two explicit-exclusion settings. NegConstraint (Xu et al. 2025) evaluates compositional constraints over a ranked corpus; ExcluIR (Zhang et al. 2025) evaluates naturally phrased exclusions using desired–excluded document pairs.

TRIX improves both metrics for every matched backbone on both benchmarks. On NegConstraint, MAP increases by 8.5–12.8 points and nDCG@10 by 7.0–9.9 points. On ExcluIR, MRR increases by 0.8–4.8 points and RR by 16.7–26.2 points. These consistent matched gains show that the same formulation operates across different retriever geometries and exclusion protocols.

The published rows provide system-level context. BGE-large with TRIX exceeds the separately reported DEO result by 3.3 MAP and 3.0 nDCG@10 points on NegConstraint. On ExcluIR, it reaches 82.2 RR, compared with 78.0 for SEAL, which uses benchmark-specific training. The matched signal-level comparison in Appendix C examines residual correction under the two evaluation protocols.

4.4 Ablation Study

Table 4 tests three questions: whether the affirmative view adds useful relevance evidence, whether residualization improves the exclusion signal, and whether attenuation and direct suppression are both needed in the final score. Here p , r , h , and g denote the positive score, residual exclusion score, thresholded exclusion penalty, and attenuation factor defined

Table 2: FollowIR results. Published instruction-trained results follow their source papers (Weller et al. 2025a,c; Gao et al. 2026). Each lower-block TRIX row is matched to the backbone above it. Score averages MAP on Core17 and Robust04 with nDCG on News21. Values are percentages ($\times 100$).

Method	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
<i>Instruction-trained retrievers</i>								
FollowIR-7B	20.0	+16.5	24.8	+13.7	29.6	+6.3	24.8	+12.2
Promptriever	21.6	+15.4	28.3	+11.7	28.5	+6.4	26.1	+11.2
InstEmb (Llama-3-8B)	24.0	+22.4	29.2	+19.1	32.3	+5.4	28.5	+15.6
<i>Matched retrieval-backbone comparisons</i>								
BM25	10.8	+3.7	13.8	−0.5	21.0	0.0	15.2	+1.1
+ TRIX	13.2	+18.4	14.0	+13.8	24.8	+20.3	17.3	+17.5
E5-Mistral	20.4	−0.3	27.9	−6.9	30.1	−0.9	26.1	−2.7
+ TRIX	24.4	+5.7	26.7	+19.6	32.8	+11.1	28.0	+12.1
BGE-large	16.8	−2.0	19.4	−5.2	21.4	+0.5	19.2	−2.2
+ TRIX	21.2	+4.3	19.8	+9.0	25.1	+6.3	22.0	+6.5
RepLLaMA	23.2	+1.1	25.7	−9.1	22.5	−1.8	23.8	−3.3
+ TRIX	25.6	+16.2	27.9	+14.1	31.8	+6.8	28.4	+12.3

Table 3: Transfer across exclusion-retrieval benchmarks. Each +*TRIX* row is matched to the Base backbone above it with the same candidates and document representations. Published DEO and SEAL values follow Lee et al. (2026b) and Zhang et al. (2025); SEAL uses ExcluIR training, while TRIX uses only its evaluation set. BGE-S/L denote BGE-small/large-en-v1.5. Values are percentages ($\times 100$).

Backbone / Method	NegConstraint		ExcluIR	
	MAP	nDCG@10	MRR	RR
<i>Published system-level references</i>				
DEO (BGE-L)	73.3	78.8	—	—
SEAL + ExcluIR training	—	—	78.4	78.0
<i>Matched frozen-backbone comparisons</i>				
BGE-S	68.4	75.2	75.0	60.1
+ <i>TRIX</i>	76.9	82.3	77.1	81.3
BGE-L	63.8	71.9	73.4	56.0
+ <i>TRIX</i>	76.6	81.8	78.2	82.2
BGE-M3	64.3	74.1	78.9	67.0
+ <i>TRIX</i>	74.7	81.1	79.7	83.7

in Section 3. All variants use the same candidates, generated views, frozen encoder, templates, embeddings, and hyperparameters.

Table 4: Controlled mechanism ablation on FollowIR. The three blocks isolate evidence construction, exclusion-signal construction, and the two operations in TRIX score composition. Score averages MAP on Core17 and Robust04 with nDCG on News21; all values are percentages ($\times 100$).

Variant	Decision score	Score	p-MRR
<i>Evidence construction</i>			
Full-query baseline	z_{full}	23.8	−3.3
Positive-view scoring	$z_{\text{full}} + p$	25.5	−0.7
<i>Exclusion-signal construction</i>			
Raw exclusion subtraction	$z_{\text{full}} + p - z_{\text{neg}}$	24.0	+14.9
Residual exclusion subtraction	$z_{\text{full}} + p - r$	26.6	+12.7
<i>TRIX score composition</i>			
TRIX w/o reward attenuation	$z_{\text{full}} + p - h$	26.3	10.8
TRIX w/o exclusion penalty	$z_{\text{full}} + pg$	26.4	8.9
Full TRIX	$z_{\text{full}} + pg - h$	28.4	+12.3

The focused positive view raises Score from 23.8 to 25.5 and p-MRR from −3.3 to −0.7, confirming that it contributes evidence beyond the full instruction. Under the same subtraction rule, replacing raw exclusion similarity with the residual recovers 2.6 Score points while retaining strongly positive p-MRR (12.7 versus 14.9). Residualization therefore reduces the relevance loss caused by direct exclusion subtraction, with a modest decrease in instruction sensitivity.

The final block tests how the residual enters the ranking score. Full TRIX improves over the variant without reward attenuation by 2.1 Score and 1.5 p-MRR points, showing that strong residual evidence should also reduce the positive contribution. Adding the explicit penalty to the attenuation-only variant yields gains of 2.0 Score and 3.4 p-MRR points. Together, the two comparisons support the asymmetric combination of attenuating affirmative support and directly suppressing strong exclusion matches.

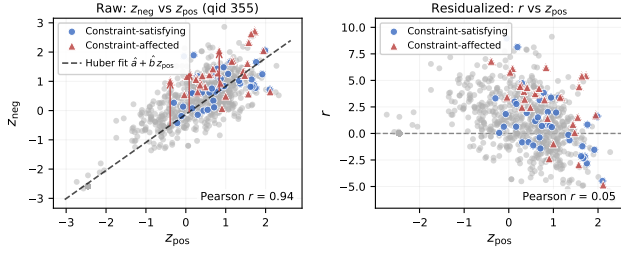


Figure 2: Residualization for an illustrative query. Left: the Huber relation between affirmative and exclusion scores. Right: standardized deviations after removing the fitted component. Positive residuals identify candidates with more exclusion similarity than predicted by their affirmative score.

4.5 Mechanism Analysis

Residual Signal. Figure 2 illustrates why raw exclusion similarity is an imprecise suppression signal. For this query, z_{neg} rises with z_{pos} , so direct subtraction also penalizes the exclusion score predicted by affirmative relevance. The fitted Huber relation estimates this shared component, and the standardized deviation r retains candidates whose exclusion similarity is unusually high relative to their affirmative score.

The raw-versus-residual comparison in Table 4 tests the ranking effect of this change; the cross-benchmark comparison is reported in Appendix C.

Effects on Ranking. TRIX converts the residual into a document-level score adjustment. Its effect relative to positive-view scoring and the full-query baseline is

$$\begin{aligned}\Delta_{\text{pos}}(d) &= S_{\text{final}}(d) - (z_{\text{full}}(d) + p(d)) \\ &= p(d)(g(d) - 1) - h(d),\end{aligned}\quad (8)$$

$$\Delta_{\text{base}}(d) = S_{\text{final}}(d) - z_{\text{full}}(d) = p(d)g(d) - h(d).$$

When $r(d) \leq \lambda$, $h(d) = 0$ and $g(d) = 1$, so $\Delta_{\text{pos}}(d) = 0$ and $\Delta_{\text{base}}(d) = p(d) \geq 0$: the positive contribution is preserved. Above the boundary, $\Delta_{\text{pos}}(d) < 0$; the positive contribution is attenuated as the exclusion penalty increases.

Figure 3 shows the resulting rank changes. The documents that remain relevant after the instruction change tend to move above the diagonal, while excluded documents tend to move below it. The separation is clearest among the highest-ranked documents, where changes have the largest effect on retrieval metrics. This behavior agrees with Eq. 8: the affirmative term promotes relevant documents, while attenuation and the explicit penalty suppress strong exclusion matches.

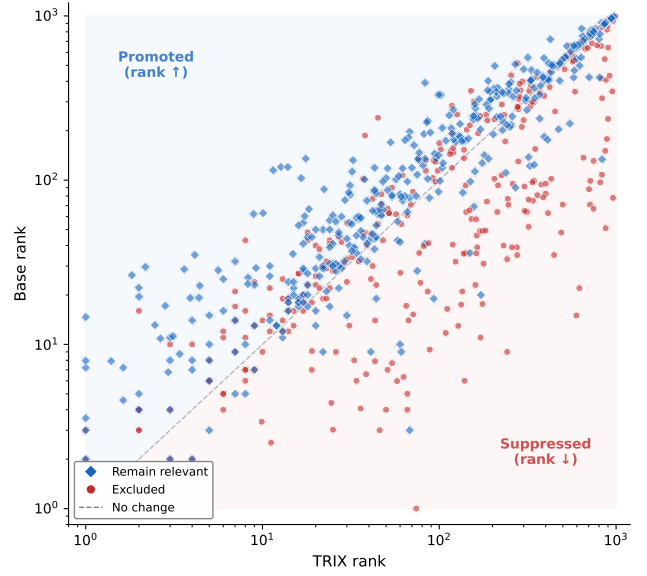


Figure 3: Base rank versus TRIX rank for documents from FollowIR exclusion queries. Each point is a document; diamonds denote documents that remain relevant after the exclusion change, and circles denote documents excluded by the new instruction. Points above the diagonal are promoted by TRIX; points below it are suppressed.

Overall, the experiments show that TRIX improves relevance and instruction sensitivity across sparse and dense retrievers, transfers to two explicit-exclusion benchmarks, and benefits from both residual signal construction and asymmetric score composition. Estimator alternatives, sensitivity, paired uncertainty, and latency are reported in Appendix B.

5 Conclusion and Limitations

This paper presented TRIX, a training-free reranking method that corrects the scores of a frozen retriever to better follow exclusion instructions. Across three benchmarks and four retriever families, TRIX consistently improves instruction-sensitive ranking without retraining or document re-encoding.

TRIX relies on two conditions: the LLM decomposition must expose useful affirmative and exclusion views, and the initial retriever must provide adequate candidate coverage. Its local affine correction may be less effective when these views remain tightly coupled in the retriever’s score space. These conditions define the intended scope of score-level correction and motivate future combinations with stronger decomposition and explicit constraint reasoning.

A Extended FollowIR Results

Table 5 gives the per-collection results for the RepLLaMA control, three representative rewriting or expansion pipelines, and TRIX. All rows use the same RepLLaMA document representations, templates, and evaluation protocol; each rewriting pipeline retains its intended query and candidate-generation procedure.

Table 5: Representative per-collection FollowIR results with RepLLaMA. Rewriting pipelines retain their intended query and candidate-generation procedures. All metrics are percentages ($\times 100$).

Method	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
<i>Matched RepLLaMA evaluation</i>								
RepLLaMA	23.2	+1.1	25.7	-9.1	22.5	-1.8	23.8	-3.3
HyDE	23.6	+6.5	24.9	-2.9	21.4	+0.7	23.4	+1.4
Query2Doc	25.9	+8.0	27.9	-7.5	24.9	-3.8	26.2	-1.1
ConvSearch-R1	22.3	+3.0	25.9	-2.5	22.0	+0.2	23.4	+0.2
TRIX	25.6	+16.2	27.9	+14.1	31.8	+6.8	28.4	+12.3

B Additional Ablations and Sensitivity

The main text isolates the positive view, residual exclusion signal, reward attenuation, and explicit penalty. The following analyses test whether the residual estimator is robust to reasonable alternatives, whether the shared configuration is stable, and whether the central paired differences are consistent with the aggregate differences reported in the main ablation.

B.1 Estimator and Normalization Choices

Table 6 audits the robust estimator, normalization, and residual construction used to obtain r .

Table 7 reports whether the shared scoring hyperparameters and candidate depth change the main conclusion, together with the end-to-end overhead required by decomposition and reranking.

B.2 Statistical Reliability

Because the controlled variants are paired at the query level, we report their paired differences in Table 8. All differences are computed per query and then averaged, so the reported Δ values reflect the mean per-query change rather than the difference of collection-level aggregates.

C Additional Exclusion-Benchmark Results

Table 3 (main text, Section 4.3) reports the complete TRIX method across NegConstraint and ExcluIR. To determine whether transfer depends on using the per-query regression residual rather than raw exclusion similarity, Table 9 applies the same matched internal comparison to both benchmarks. All rows use the same frozen retriever, candidate set, decomposition, query templates, document embeddings, and shared λ, τ ; the raw and residual correction rows differ only in the exclusion signal supplied to the same TRIX scoring function.

Under the same TRIX scoring rule, the regression residual provides a stronger effectiveness-exclusion profile than raw similarity on both benchmarks. This comparison supports transfer of the ranking signal, not recovery of the underlying logical semantics.

D Qualitative Analysis

The qualitative analysis contains one representative success and one semantic-entanglement failure from FollowIR

(Core17) with RepLLaMA. Each case reports the benchmark, query ID, complete request, decomposed views, candidate scores, and the ranking change. The success case is the query with the largest single-document rank drop; the failure case is the query with the smallest rank change among those carrying an exclusion instruction.

E Decomposition and Reproducibility Details

The instruction decomposer is the frozen Qwen3-4B model (Yang et al. 2025). It receives the complete user request and returns structured affirmative and exclusion content in one generation. The full view Q_{full} is the original request itself, used unchanged for candidate generation; Q_{pos} retains the topic and affirmative conditions while removing exclusion content, and Q_{neg} contains the excluded content, or [NONE] when no exclusion is present. We use the fixed TSC_BALANCED_t01 prompt with temperature 0.1, top- $p = 0.9$, at most 512 new tokens, and thinking disabled. Malformed or missing fields are mapped to the corresponding empty view. The system prompt is:

```
You are an expert IR query optimizer for Dense
Vector Models. Decouple "Original Query" and
"Instruction" into Q_plus and Q_minus.
```

```
Q_plus rules: (1) Fluent natural language phrase,
not a keyword list. (2) Semantic topic only;
remove format noise ("articles", "documents",
"reports"). (3) Remove conversational connectors
("such as", "including"). (4) Sharp, coherent.
```

```
Q_minus rules: (1) Extract only critical
high-level exclusion concepts. (2) Concise
natural phrasing. (3) If no exclusion,
output "[NONE]".
```

```
No-negation rule: DENSE RETRIEVERS IGNORE LOGIC
WORDS. NEVER use negation in Q_minus ("no",
"not", "non-", "without"). Convert negations to
AFFIRMATIVE entities:
```

```
BAD: "non-United Kingdom"
-> GOOD: "United States, France, global"
BAD: "violence outside Ireland"
-> GOOD: "violence in England, international"
```

```
Output ONLY exact entities to exclude.
```

```
Output JSON:
```

```
{
  "Reasoning_Steps": "1. Core topic. 2. Filter
noise. 3. Exclusions (negations->affirm).",
```

Table 6: Secondary TRIX design choices on FollowIR. Each row changes one estimation or normalization choice while retaining the controlled candidate set and three evidence views.

Variant	Change from TRIX	Score	p-MRR
TRIX	None	28.4	+12.3
OLS fit	Replace Huber regression	24.9	+10.6
Mean/std scaling	Replace median/MAD scaling	24.8	+8.5
Uncentered residual	Omit residual recentering	24.8	+10.8

Table 7: Sensitivity and efficiency. Score and p-MRR are averaged over FollowIR; latency is measured per query under a fixed hardware setting.

Analysis	Setting	Avg Score	Avg p-MRR
$\lambda = 0.5$	default	28.4	+12.3
$\lambda = 1.0$		27.6	+10.8
$\lambda = 1.5$		26.9	+9.2
$\lambda = 2.0$		25.8	+7.5
$\tau = 0.1$	default	28.4	+12.3
$\tau = 0.2$		28.2	+11.9
$\tau = 0.5$		27.8	+10.4
$\tau = 1.0$		27.1	+8.7
$K = 10$	candidate depth	26.3	+18.1
$K = 50$		27.9	+20.8
$K = 100$	default	28.5	+22.5
$K = 200$		28.4	+22.1
Decomposition	One structured LLM call	0.3 s / query	
TRIX reranking	Cached document embeddings	5 ms / query	

Table 8: Paired differences for the central component comparisons on FollowIR.

Matched comparison	Δ Score	Δ p-MRR
Positive-view scoring vs. full-query baseline	+1.7	+2.6
Residual vs. raw exclusion subtraction	+2.6	−2.2
Full TRIX vs. TRIX w/o reward attenuation	+2.1	+1.5
Full TRIX vs. TRIX w/o exclusion penalty	+2.0	+3.4

Table 9: Matched exclusion-signal ablation across exclusion benchmarks with BGE-large-en-v1.5.

Variant	NegConstraint		ExcluIR	
	MAP	nDCG@10	MRR	RR
Full-query baseline	63.8	71.9	73.4	56.0
Positive-view scoring	76.0	81.8	75.0	58.0
Raw exclusion with TRIX scoring	76.6	81.8	76.0	60.0
Residual exclusion with TRIX scoring	76.6	81.8	78.2	82.2

```

"Q_plus": "Fluent natural language query",
"Q_minus": "Core exclusions or [NONE]"
}

```

The user prompt template is:

```

Decouple into Q_plus and Q_minus:
Query: "{query}"
Instruction: "{instruction}"

```

All document representations are computed once with the base retriever’s document template, and all three query views use its query template. For numerical stability, Huber residuals and channel scores use a 10^{-6} floor in their median/MAD normalization. The regression is fit independently on each candidate set and uses no relevance labels. The scoring hyperparameters λ and τ are fixed before evaluation and shared across datasets and retrievers; no query-specific tuned coefficients or metric-based tuning are introduced. The released configuration records these constants and the candidate-set depth for each experiment.

Table 10: Representative TRIX cases from FollowIR Core17 with RepLLaMA. Each case reports the operational views, score-space evidence, and ranking change under the complete scoring rule.

Outcome	Request and views	Candidate evidence	Ranking effect and interpretation
Success	Benchmark: FollowIR (Core17), QID 355 Query: “Identify documents discussing the development and application of space-borne ocean remote sensing” Exclusion: land-bound sciences, agriculture, forestry, marketing, temperature Views: Q_{full} (full query), Q_{pos} (ocean remote sensing in oceanography, seabed prospecting, marine sciences), Q_{neg} (land-bound sciences, agriculture, forestry, marketing, temperature)	Original top-1 doc ($s_{full} = 0.725$) scores high on both s_{pos} and s_{neg} channels (agriculture/forestry content). Residual r separates the exclusion signal from shared marine-science relevance. After TRIX: top-1 score rises to 1.907 for ocean-only doc; land-bound doc drops to rank 580.	Land-bound/agriculture documents drop from top-10 to rank > 500 . Ocean remote-sensing documents rise to top-5. Interpretation: Residual penalty correctly identifies exclusion signal when exclusion and target content are topically distinct.
Failure	Benchmark: FollowIR (Core17), QID 310 Query: “Evidence that radio waves from radio towers or car (mobile) phones affect brain cancer occurrence” Exclusion: leukemia Views: Q_{full} (full query), Q_{pos} (radio waves and brain cancer incidence), Q_{neg} (leukemia)	Original top-1 doc ($s_{full} = 0.795$) discusses brain cancer broadly. s_{neg} channel for “leukemia” is low because leukemia-specific mentions are rare in the candidate set, yielding near-zero residual r for the top document. After TRIX: top-1 score rises to 1.689 but ranking is unchanged.	Top-5 ranking is unchanged; excluded-topic documents remain in place. Interpretation: When the exclusion term rarely co-occurs with the target in the candidate set, the residual is too small to trigger a penalty—the failure mode is low negative-channel signal rather than semantic entanglement.

References

- Asai, A.; Schick, T.; Lewis, P.; Chen, X.; Izacard, G.; Riedel, S.; Hajishirzi, H.; and Yih, W. 2023. Task-aware Retrieval with Instructions. In Rogers, A.; Boyd-Graber, J. L.; and Okazaki, N., eds., *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, volume ACL 2023 of *Findings of ACL*, 3650–3675. Association for Computational Linguistics.
- Cai, Z.; Tao, C.; Shen, T.; Xu, C.; Geng, X.; Lin, X. A.; He, L.; and Jiang, D. 2023. HypeR: Multitask Hyper-Prompted Training Enables Large-Scale Retrieval Generalization. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Gao, L.; Ma, X.; Lin, J.; and Callan, J. 2023a. Precise Zero-Shot Dense Retrieval without Relevance Labels. *arXiv preprint arXiv:2212.10496*.
- Gao, L.; Ma, X.; Lin, J.; and Callan, J. 2023b. Precise Zero-Shot Dense Retrieval without Relevance Labels. In *ACL*, 1762–1777.
- Gao, T.; Fang, J.; Zhang, X.; Liu, Z.; Liu, C.; Liu, P.; and Jiang, Q. 2026. InstEmb: Instruction-Following Embeddings through Glimpses of the Future. In *Proceedings of the 43rd International Conference on Machine Learning*.
- Hoveyda, M.; Piepenbrock, J.; de Vries, A. P.; de Rijke, M.; and Hasibi, F. 2026. OrLog: Resolving Complex Queries with LLMs and Probabilistic Reasoning. In Campos, R.; Jatowt, A.; Lan, Y.; Aliannejadi, M.; Bauer, C.; MacAvaney, S.; Anand, A.; Ren, Z.; Verberne, S.; Bai, N.; and Mansoury, M., eds., *Advances in Information Retrieval - 48th European Conference on Information Retrieval, ECIR 2026, Delft, The Netherlands, March 29 - April 2, 2026, Proceedings, Part I*, volume 16483 of *Lecture Notes in Computer Science*, 98–114. Springer.
- Huber, P. J. 1964. Robust Estimation of a Location Parameter. *The Annals of Mathematical Statistics*, 35(1): 73–101.
- Jiang, P.; Lin, J.; Cao, L.; Tian, R.; Kang, S.; Wang, Z.; Sun, J.; and Han, J. 2025. DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning. *CoRR*, abs/2503.00223.
- Lee, T.; Park, J.; Hwang, S.; and Jang, J. 2026a. DEO: Training-Free Direct Embedding Optimization for Negation-Aware Retrieval. *arXiv preprint arXiv:2603.09185*.
- Lee, T.; Park, J.; Hwang, S.; and Jang, J. 2026b. DEO: Training-Free Direct Embedding Optimization for Negation-Aware Retrieval. *CoRR*, abs/2603.09185.
- Li, X.; et al. 2025. DeepRetrieval: Retrieval-Augmented Query Generation for Information Retrieval. *arXiv preprint*.
- Ma, X.; Wang, L.; Yang, N.; Wei, F.; and Lin, J. 2024. Fine-Tuning LLaMA for Multi-Stage Text Retrieval. In *SIGIR*, 2421–2425.
- Pokhrel, A.; and Sheshappanavar, S. V. 2026. Negation Matters: Training-Free Negation-Aware Image Retrieval. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 11353–11362.
- Raudaschl, S. 2024. RAG-Fusion: A New Take on Retrieval-Augmented Generation.
- Robertson, S. E.; and Zaragoza, H. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 3(4): 333–389.
- Rousseeuw, P. J.; and Croux, C. 1993. Alternatives to the Median Absolute Deviation. *Journal of the American Statistical Association*, 88(424): 1273–1283.

- Singh, R.; Kumar, R.; and Sridhar, V. 2023. NLMs: Augmenting Negation in Language Models. In Bouamor, H.; Pino, J.; and Bali, K., eds., *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, volume EMNLP 2023 of *Findings of ACL*, 13104–13116. Association for Computational Linguistics.
- Su, H.; Shi, W.; Kasai, J.; Wang, Y.; Hu, Y.; Ostendorf, M.; Yih, W.; Smith, N. A.; Zettlemoyer, L.; and Yu, T. 2023. One Embedder, Any Task: Instruction-Finetuned Text Embeddings. In Rogers, A.; Boyd-Graber, J. L.; and Okazaki, N., eds., *Findings of the Association for Computational Linguistics: ACL 2023, Toronto, Canada, July 9-14, 2023*, volume ACL 2023 of *Findings of ACL*, 1102–1121. Association for Computational Linguistics.
- Wang, L.; Yang, N.; Huang, X.; Yang, L.; Majumder, R.; and Wei, F. 2024. Improving Text Embeddings with Large Language Models. In *ACL*, 11897–11916.
- Wang, L.; Yang, N.; and Wei, F. 2023a. Query2doc: Query Expansion with Large Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 9414–9423. Association for Computational Linguistics.
- Wang, L.; Yang, N.; and Wei, F. 2023b. Query2doc: Query Expansion with Large Language Models. In *EMNLP*, 9414–9423.
- Weller, O.; Chang, B.; MacAvaney, S.; Lo, K.; Cohan, A.; Durme, B. V.; Lawrie, D.; and Soldaini, L. 2025a. FollowIR: Evaluating and Teaching Information Retrieval Models to Follow Instructions. In *NAACL*, 11926–11942.
- Weller, O.; Chang, B.; MacAvaney, S.; Lo, K.; Cohan, A.; Van Durme, B.; Lawrie, D.; and Soldaini, L. 2025b. FollowIR: Evaluating and Teaching Information Retrieval Models to Follow Instructions. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 11926–11942. Albuquerque, New Mexico: Association for Computational Linguistics.
- Weller, O.; Durme, B. V.; Lawrie, D.; Paranjape, A.; Zhang, Y.; and Hessel, J. 2025c. Promptriever: Instruction-Trained Retrievers Can Be Prompted Like Language Models. In *ICLR*.
- Weller, O.; Lawrie, D. J.; and Durme, B. V. 2024. NevIR: Negation in Neural Information Retrieval. In Graham, Y.; and Purver, M., eds., *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics, EACL 2024 - Volume 1: Long Papers, St. Julian's, Malta, March 17-22, 2024*, 2274–2287. Association for Computational Linguistics.
- Weller, O.; Van Durme, B.; Lawrie, D.; Paranjape, A.; Zhang, Y.; and Hessel, J. 2025d. Promptriever: Instruction-Trained Retrievers Can Be Prompted Like Language Models. In *International Conference on Learning Representations*.
- Xiao, S.; Liu, Z.; Zhang, P.; Muennighoff, N.; Lian, D.; and Nie, J. 2024. C-Pack: Packed Resources For General Chinese Embeddings. In *SIGIR*, 641–649.
- Xu, G.; Zhang, Z.; Mei, W.; Liang, J.; Lu, W.; Zhang, X.; Yang, Z.; Ma, X.; Xiao, Y.; and Yang, D. 2025. Logical Consistency is Vital: Neural-Symbolic Information Retrieval for Negative-Constraint Queries. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Findings of the Association for Computational Linguistics: ACL 2025*, 1828–1847. Vienna, Austria: Association for Computational Linguistics.
- Yang, A.; Li, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Gao, C.; Huang, C.; Lv, C.; et al. 2025. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*.
- Yin, B.; Lu, X.; Qi, Z.; and Shen, X. 2026. When Should Queries Be Decomposed? A Stage-Aware Study of Query Decomposition for Multi-Condition Retrieval. *arXiv preprint arXiv:2606.08577*.
- Zeng, Q.; et al. 2026. Dual-View Training for Instruction-Following Information Retrieval. *arXiv preprint arXiv:2604.18845*.
- Zhang, W.; Zhang, M.; Wu, S.; Pei, J.; Ren, Z.; de Rijke, M.; Chen, Z.; and Ren, P. 2025. ExcluIR: Exclusionary Neural Information Retrieval. In *AAAI*, 13295–13303.
- Zhuang, Y.; Trinh, A.; Qiang, R.; Sun, H.; Zhang, C.; Dai, H.; and Dai, B. 2025. Towards Better Instruction Following Retrieval Models. *arXiv preprint arXiv:2505.21439*.